so the integral of the probability density is finite. And, importantly, it has no nodes. So it has a chance of being our ground state.

Let's see if we can figure out what the Hamiltonian does to this function. The first term of the Hamiltonian (the left side of Eq. 10.14) tells us to apply the operator

$$-\frac{\hbar^2}{2}\frac{\partial}{\partial x^2}$$

to $\psi(x)$. Let's calculate that term, one derivative at a time. The first step is

$$\frac{\partial\psi(x)}{\partial x} = -\frac{\omega}{2\hbar}(2x)e^{-\frac{\omega}{2\hbar}x^2},$$

which simplifies to

$$\frac{\partial\psi(x)}{\partial x} = -\frac{\omega}{\hbar}xe^{-\frac{\omega}{2\hbar}x^2}.$$

When we take the second derivative, there will be two terms because of the product rule:

$$\frac{\partial^2\psi(x)}{\partial x^2} = -\frac{\omega}{\hbar}e^{-\frac{\omega}{2\hbar}x^2} + \frac{\omega^2}{\hbar^2}x^2e^{-\frac{\omega}{2\hbar}x^2}.$$

Let's plug this result back into Eq. 10.14, and at the same time replace ψ on the right side with our guess, $e^{-\frac{\omega}{2\hbar}x^2}$:

$$\frac{\hbar}{2}\omega e^{-\frac{\omega}{2\hbar}x^2} - \frac{1}{2}\omega^2x^2e^{-\frac{\omega}{2\hbar}x^2} + \frac{1}{2}\omega^2x^2e^{-\frac{\omega^2}{2\hbar}x^2} = Ee^{-\frac{\omega}{2\hbar}x^2}.$$